

Idea F: The Knowledge Graph Builder Agent

An Agentic GenAI / LLM model for Japan's Shipbuilding & Green-Technology Study

1. The Idea in Simple Words

Imagine you have a big messy pile of reports about Japan's shipbuilding and green-energy plans. This AI reads them all and turns them into a simple **map of dots and lines** — and then you can **chat with that map**. It is like turning many pages of dense reports into a living mind-map you can talk to.

2. How It Works

A knowledge graph has only two things: **nodes** (the dots = real things like companies, technologies, policies, targets) and **edges** (the lines = how they relate, e.g. "acquires", "funds", "targets", "threatens"). The agent reads the documents, pulls out these dots and lines automatically, stores them, and then answers your questions by walking along the connections.

- **Read** — the agent ingests reports / PDFs (OECD, METI, Ember, Nippon.com).
- **Extract** — the LLM pulls out entities ("Imabari", "10 GW", "offshore wind").
- **Link** — the LLM figures out relationships ("acquires", "targets", "supports").
- **Build** — it stores everything in a graph database.
- **Ask** — you ask plain-English questions; the agent answers using the map.

3. Sample Knowledge Graph (from your brief)

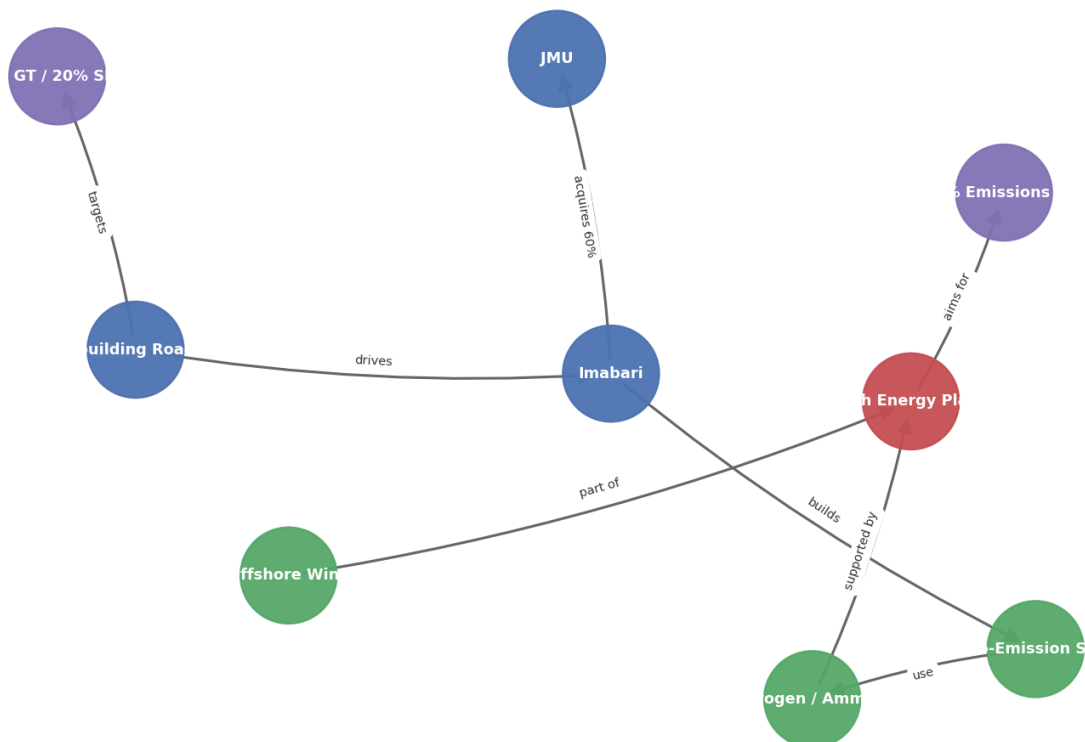


Figure 1: How shipbuilding (blue), green tech (green), policy (red) and national targets (purple) connect — visually proving the convergence thesis.

4. Why This Is Agentic AI + GenAI + LLM

Buzzword	What it actually means in this project
Agentic AI	The AI decides what to extract, merges duplicates, resolves conflicting numbers, and walks multi-step path
GenAI / LLM	The LLM reads messy text, extracts structured entities & relationships, and writes human-readable answer

5. Example: Multi-hop Question

You ask: "How does the Imabari–JMU merger connect to Japan's emissions goal?"

The agent walks the path: Imabari → acquires JMU → builds zero-emission ships → use hydrogen/ammonia → supports the 7th Energy Plan → 46% emissions cut.

It answers: "Indirectly but strongly — consolidation gives Japan the scale to mass-produce ammonia/hydrogen ships, which feed directly into the national decarbonization target."

6. Technology Stack

- **Graph database:** Neo4j (production) or NetworkX (prototype).
- **Brain:** an LLM (GPT / Claude, or self-hosted Llama / Mistral).
- **Extraction:** LLM with structured JSON output.
- **Query layer (GraphRAG):** English question → graph lookup → written answer.
- **Visualization:** PyVis / D3.js / Neo4j Bloom.

7. Minimum Viable Product

- Take the research brief as input text.
- LLM extracts entities + relationships into JSON.
- Load into NetworkX (no database setup needed).
- Draw the graph and answer 2–3 questions about it.

This MVP is exactly what the accompanying Python model implements.